



Does the Order Work?

Read the algorithm and check the order

Name:

Date:

★ I can tell if an algorithm's steps are in an order that works.

Check the order

Read each algorithm. Does the order make sense? Circle YES if it works, or NO if a step is in the wrong place. Then tell why. A step that needs another step must come AFTER it.

Algorithm A — draw a sun

1. draw a circle

2. add rays around it

3. colour it yellow

1 Does A work?

Circle YES or NO. Tell why the order does or does not work.

Algorithm B — have cereal

1. eat it

2. pour the milk

3. pour the cereal in the bowl

2 Does B work?

Circle YES or NO. Tell what is wrong with the order.

Algorithm C — go down the slide

1. climb the ladder

2. slide down

3. sit at the top

3 You try — C does not work

Circle YES or NO. Then write the 3 steps of C in an order that DOES work.

Big idea

An algorithm only works if every step comes after the steps it needs. If a step is in the wrong place, the order does not work — even if all the steps are there.



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Quick facilitation notes & answer key

What this worksheet teaches

Students read three plans and judge whether the order works, explaining why, then rewrite a broken one. Reading a plan and deciding if the order succeeds — and fixing it — is the heart of debugging. No coding experience needed.

Before you start (no computer needed)

No computer — paper and a pencil. You read each prompt aloud. Optional: cut the steps into cards so students can physically slide them into order before writing.

How to lead it (about 20 minutes)

1. Show them (you do). Read a plan aloud and ask: "does each step have what it needs?"
2. Do one together. Plan A works; Plan B does not (you cannot eat before you pour).
3. Let them try. Students judge Plan C and rewrite it so it works.

Answer key (these are the verified correct answers)

Exercise 1 — Plan A WORKS (YES): draw the circle, add rays, colour it — each step has what it needs.

Exercise 2 — Plan B does NOT work (NO): eating is first, but you must pour the cereal then the milk before you can eat; eating should be last.

Exercise 3 — Plan C does NOT work (NO). A working order: climb the ladder, sit at the top, slide down (you must sit before you slide).

If students get stuck — and ways to adjust

Watch for: saying YES because the steps look familiar. Check each step has what it needs from earlier steps.

Make it easier: act out the broken plan so the problem is obvious.

Make it harder: ask them to explain exactly which step is too early and why.

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential events.

In plain terms: students put step-by-step instructions in the right order to get a result.

C3.2 — read and alter existing code, including code that involves sequential events, and describe how changes to the code affect the outcomes.

In plain terms: students read a set of steps, change or reorder one, and explain what the change did.

You'll know it worked when

The child judges each plan correctly and rewrites the broken one into a working order.